March 10th , 2026

Integrated MV, AI and PLC

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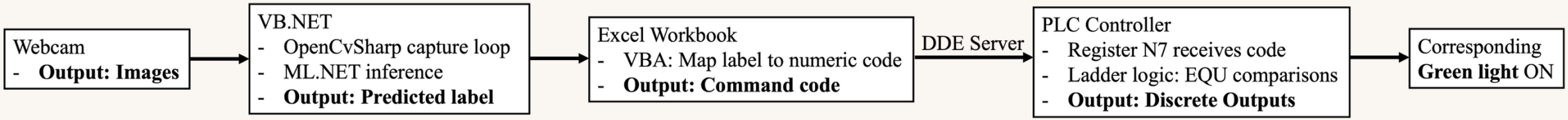
Manufacturing and Mechanical Engineering Technology

What do you wish to learn from this session?

Which area are you familiar with?

- Machine Vision
- Artificial Intelligence
- Programmable Logic Controller

Machine Vision (MV)



- **How it works?**
  - Sensors (cameras) capture images
  - Processed using algorithms (AI/ML) to detect, analyze, and measure features
  - Decision-making → Output control
- **Components:**
  - Sensors (cameras)
  - Image processing hardware/software
  - Output control devices
  - Communication
- **Applications:** In Industrial settings…
  - Robotic guidance
  - Quality Control
  - Automate visual inspection

Today's agenda

- **10:15 – 11:00 AM**
  - AI/ML
  - PLC & Ladder Logic
  - Data & Signal communication

- **11:15 – 12:00 AM**
  - Lab Demonstration

AI/ML

Which statement best describes machine learning?

A) A fixed set of rules written by the programmer

B) A method where a model learns patterns from examples to make predictions

C) A camera that automatically improves image quality

In supervised learning, what do we need?

A) Only unlabeled images

B) Labeled examples (images + correct answers like OK/NG)

C) Only a pretrained model, no data needed

Which is a classification task?

A) Predict the exact defect size in millimeters

B) Predict OK vs NG (or part type A/B/C)

C) Compress an image file

AI/ML

- **What is Machine Learning?**

➢ A program that learns patterns from examples to make predictions

Input (Data) →
Output →
Machine Learning → Rules

Input (Data) →
Rules →
Traditional Programming → Output

AI/ML

- **Types of Machine Learning**

➢ **Supervised Learning**

-Input feature vetors (Data) with target values (Labels)

➢ **Unsupervised Learning**

-Input feature vetors (Data) without target values (Labels)

➢ **Semi-supervised Learning**

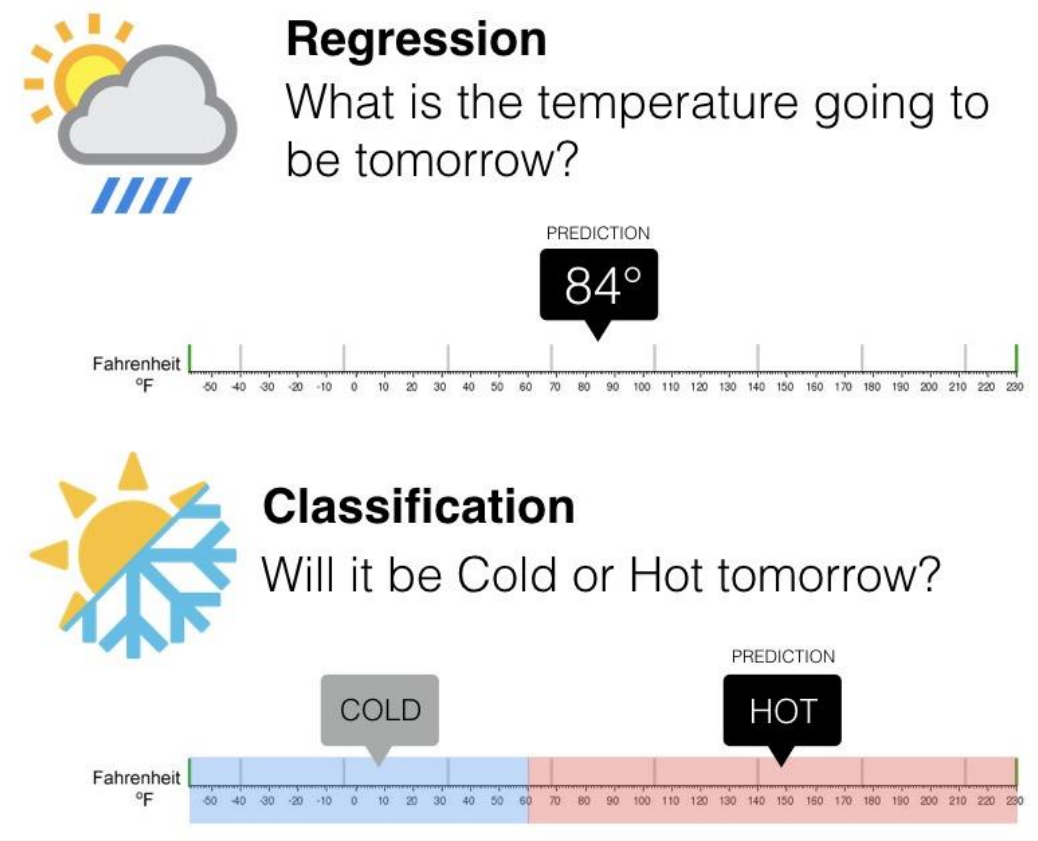
-A large amount of the data are unlabeled

➢ **Reinforcement Learning**

- Learn through trial and error by receiving rewards/penalties

AI/ML

- **Types of Machine Learning**
  - Different types of output



- Regression:

Continuous Values

- Classification:

Discrete outcomes
(Label + Confidence/probability)
Hot: 0.98, Cold: 0.02 ?
Hot: 0.52 , Cold: 0.48 ?
=> Threshold and uncertain zone

Q4. Which policy should we use when the model is unsure?

Always pick the top class, no matter what

B) Only accept the top class if probability ≥ T; otherwise mark UNCERTAIN / re-check

C) Average the probabilities and send to PLC

D) D) Ignore confidence and act on label only

AI/ML

- **Training**
  - Learn the Pattern from the data

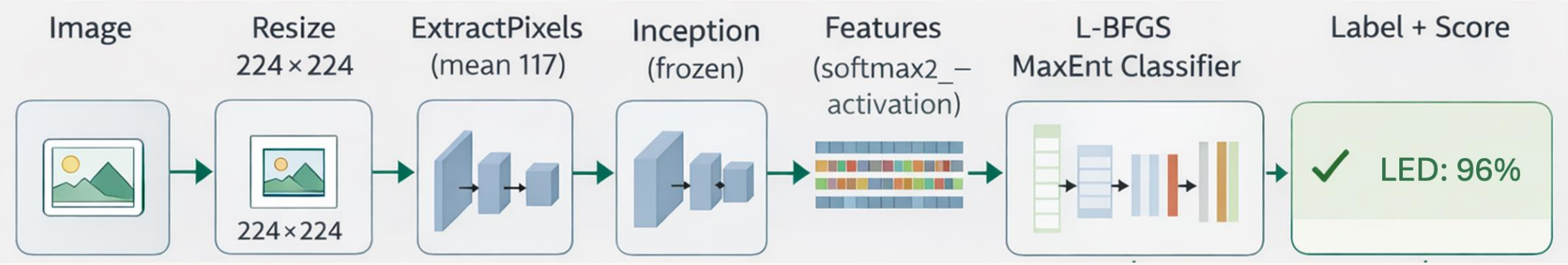
- **Validation**
  - Evaluate/ Fine-tune the trained model with new images

- **Testing**
  - Final evaluate the trained model with new images

- **Dataset Split Ratio**
  - Train: Validate: Test = 8:1:1

AI/ML

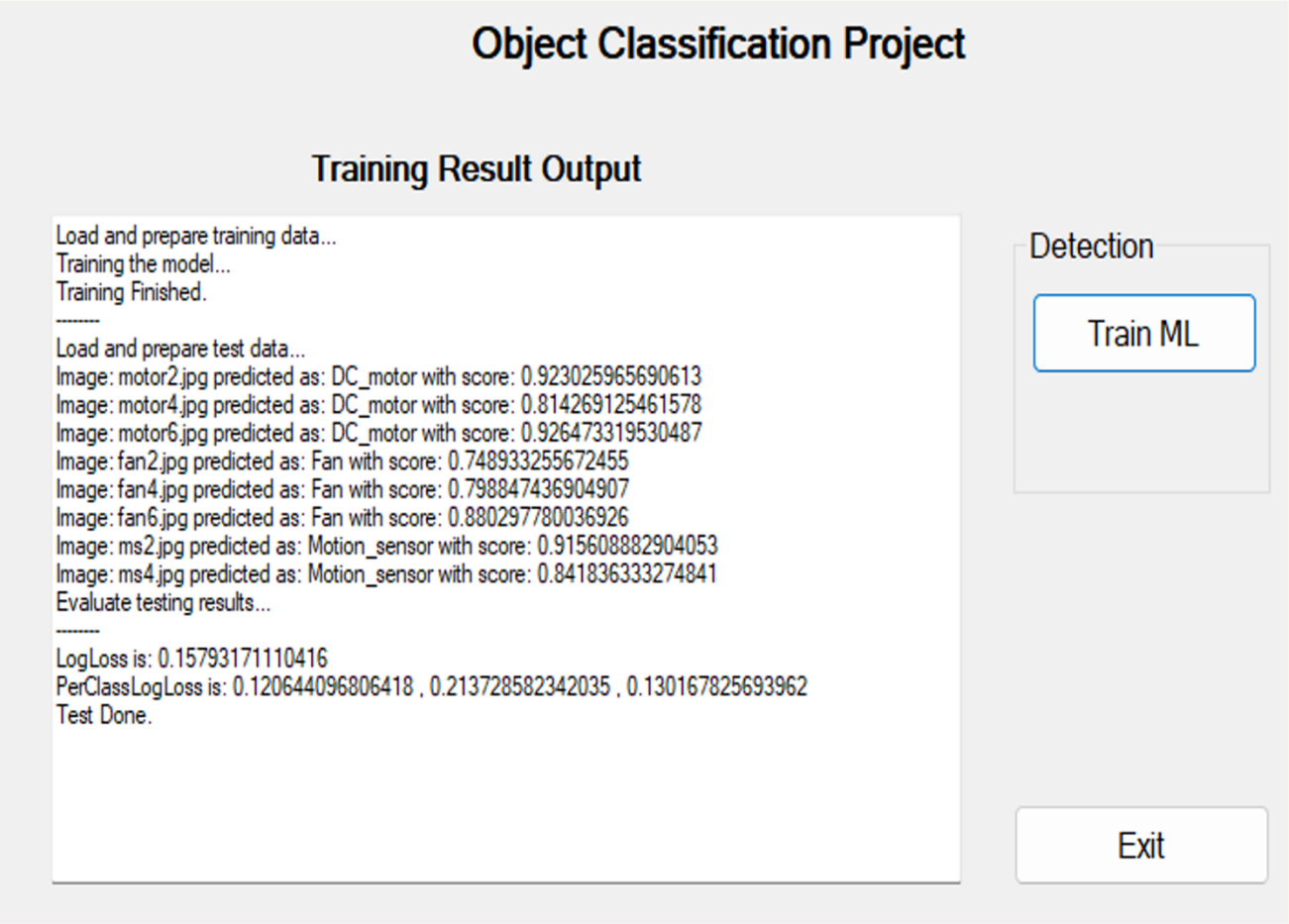
- **Role:**
  - Process the images to detect, analyze, and measure features
  - Decision-making



- **Inception "Pre-trained" Model:**
  - Transfer learning → Require small dataset → Fast training
  - Deployable on .NET/ other application platform
  - Other options: MobileNet, ResNet, YOLO

AI/ML

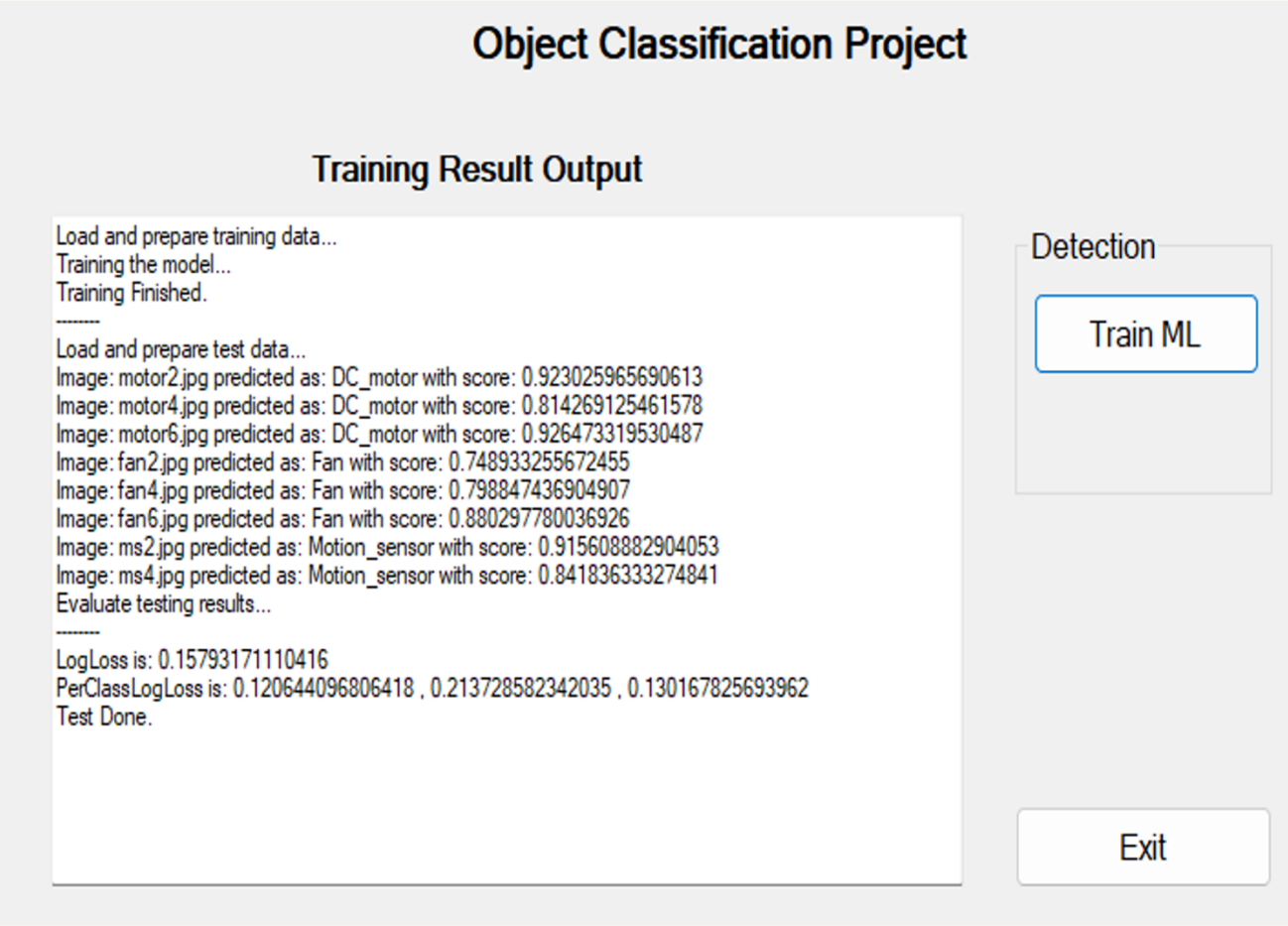
- **Lab Exercise:**
  - Design scenarios
  - Collect images for new categories
  - Train and observe the results



What are the applicable scenario ideas?

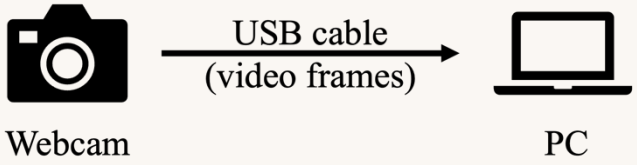
AI/ML

- **Lab Exercise:**
  - Design scenarios
  - Collect images for new categories
  - Train and observe the results
- **Variations:**
  - Effect of train-test split ratio
  - Effect of image quantity/quality
  - Effect of decision threshold setting

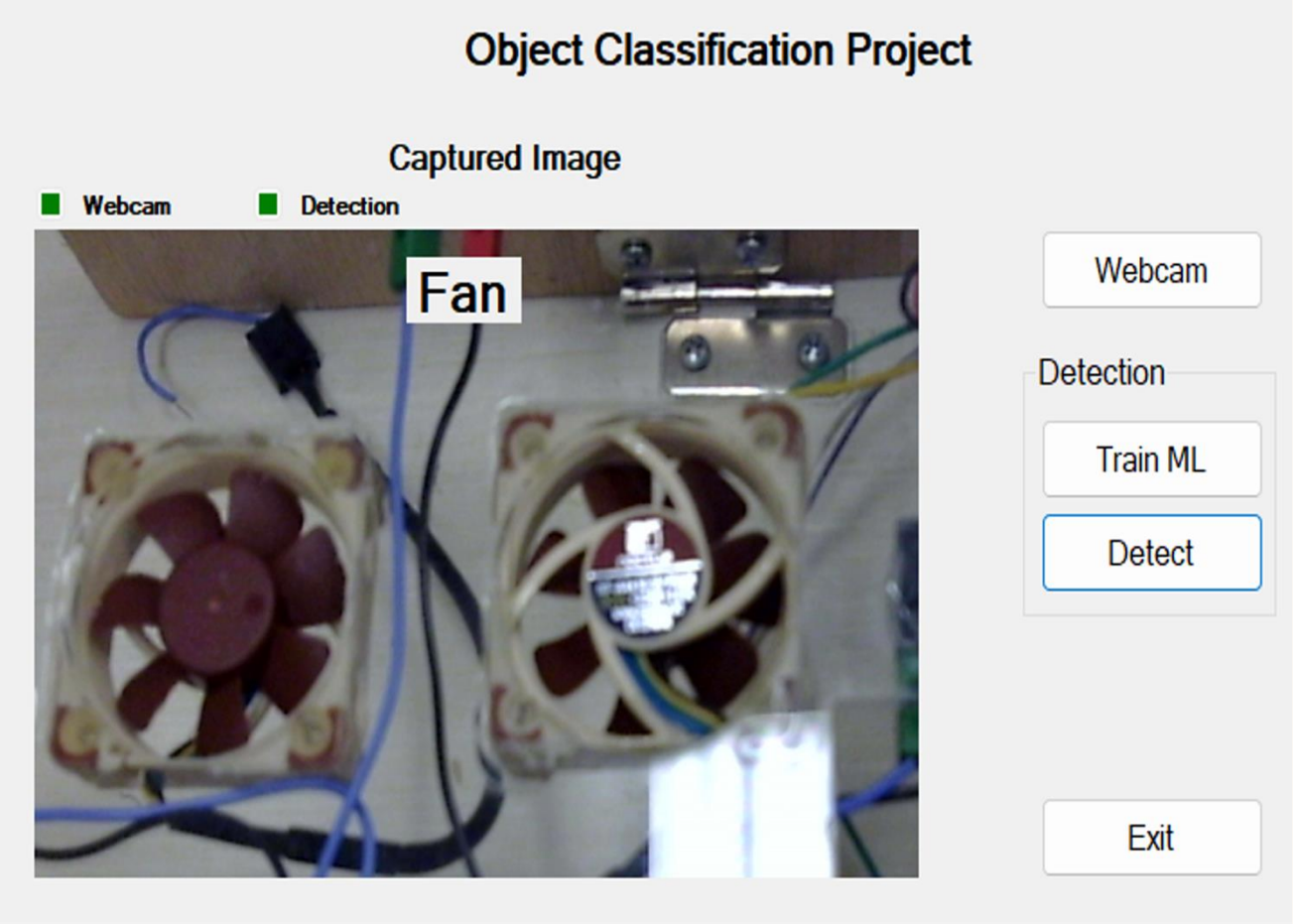


AI/ML

- **Lab Exercise:**
  - Real-time deployment
  - Integrate with Sensor devices
  - Observe and discuss the results



- **Variations:**
  - Analyze accuracy over trials
  - Effect of train-test split ratio
  - Effect of image quantity/quality
  - Effect of decision threshold setting



Q1. What does it mean that we use a “pretrained” model (like Inception)?

A) It was trained on our factory dataset already

B) It learned general visual features from a large dataset, and we reuse them

C) It guarantees perfect accuracy

Q2. Why is transfer learning a good fit for a short lab with limited data?

A) It removes the need for labeling

B) We can freeze most layers and train a small new classifier quickly

C) It prevents communication delays

Q3. If we raise the pass threshold (e.g., from 0.60 to 0.85), what usually happens?

A) More parts pass, fewer rejects

B) Fewer false accepts, but more false rejects

C) The prediction become more reliable

PLC & Ladder Logic

What does “PLC” stand for?

A) Personal Laptop Compiler

B) Production Line Coordinator

C) Programmable Logic Controller

What is the main role of a PLC in an automation system?

A) Train machine-learning models

B) Execute real-time, deterministic control of machines using inputs/outputs

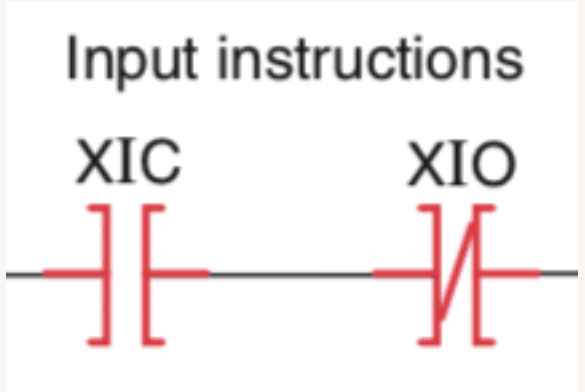
C) Store sensor readings and analyze them

In ladder logic, an "instruction" is most like...

A) A conditional check

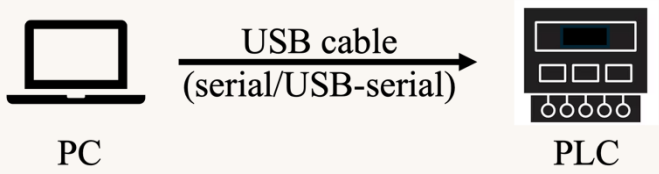
B) An output status

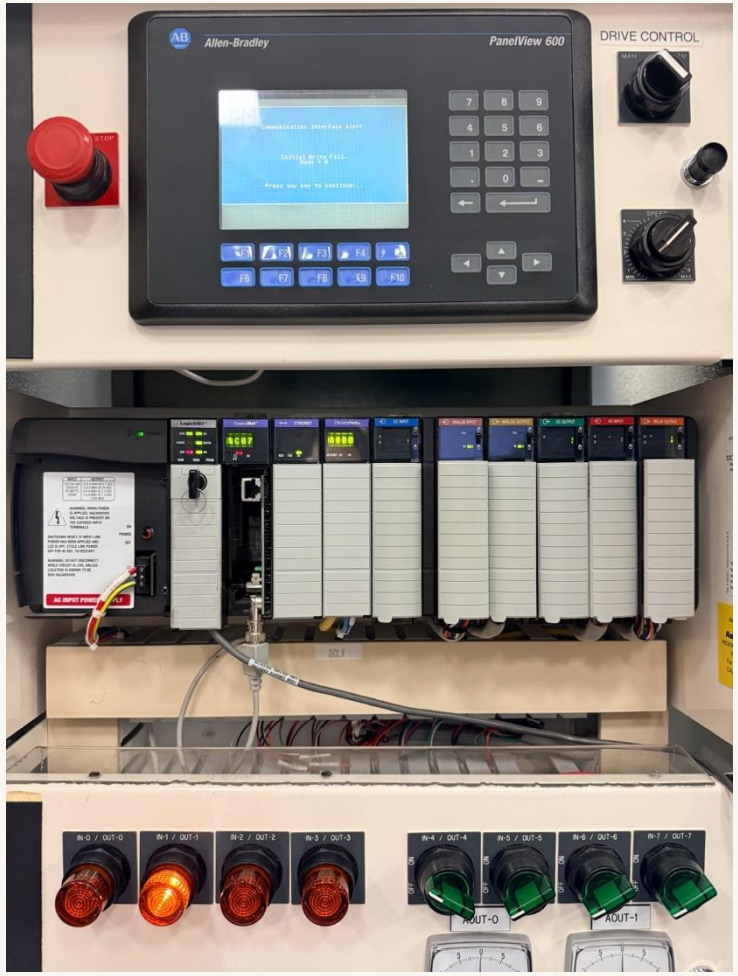
C) A database query



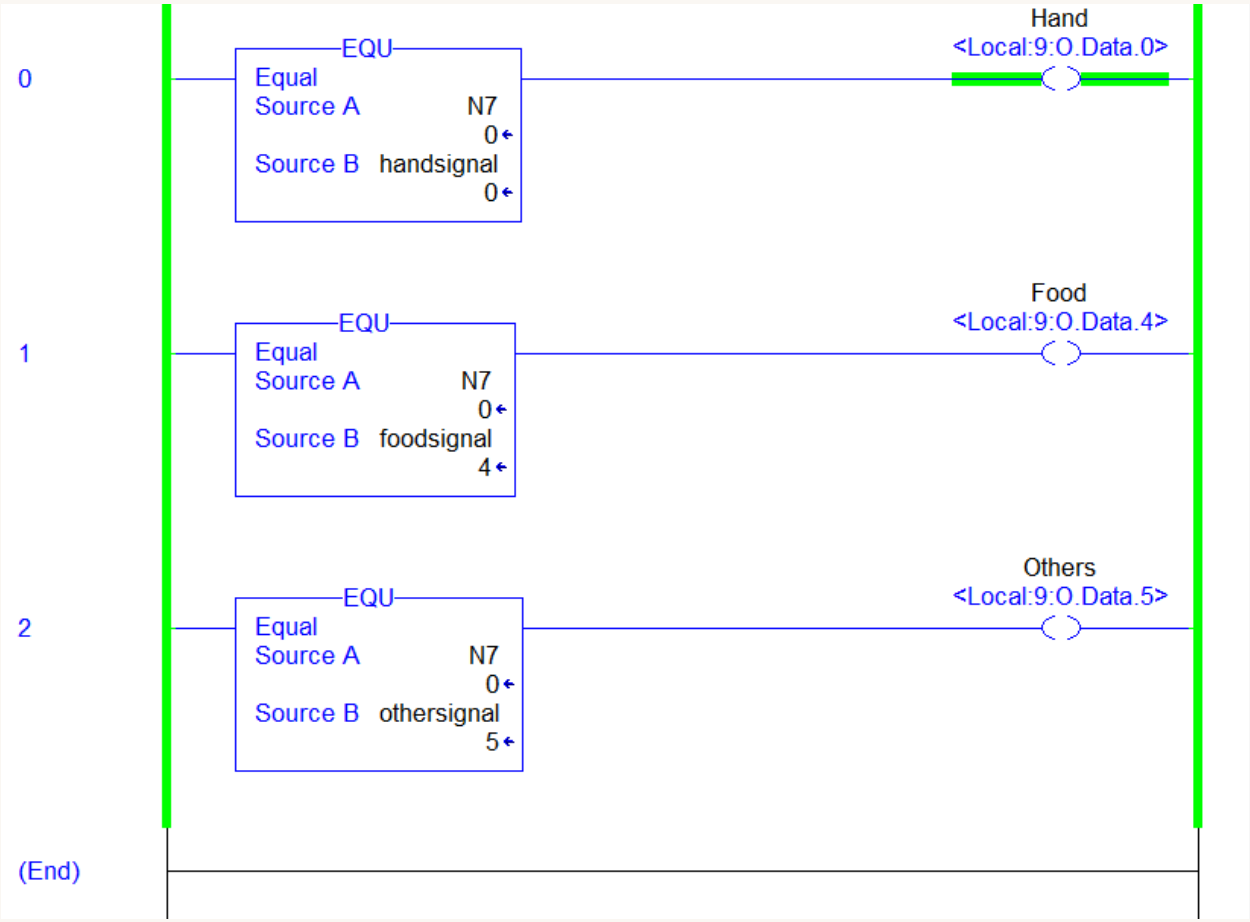
PLC & Ladder Logic

- **Role:**
  - Real-time decision “executor”
  - Applies control logic deterministically
  - Drives physical outputs





PLC & Ladder Logic



- **EQU (Equal):**
  - Compare instruction
  - "TRUE" when Source A = Source B
  - Source can be tags, constants, numbers

Name	Alias For	Base Tag	Data Type
Food	Local:9:O.Data.4(C)	Local:9:O.Data.4(C)	BOOL
N7			DINT
Others	Local:9:O.Data.5(C)	Local:9:O.Data.5(C)	BOOL
othersignal			INT
foodsignal			INT
handsignal			INT
Hand	Local:9:O.Data.0(C)	Local:9:O.Data.0(C)	BOOL

PLC & Ladder Logic

- **Outputs**
  - The address of where the outputs are connected
- **Constants**
  - Reference values to compare with
- **N7**
  - PLC integer register
  - Incoming data mailbox

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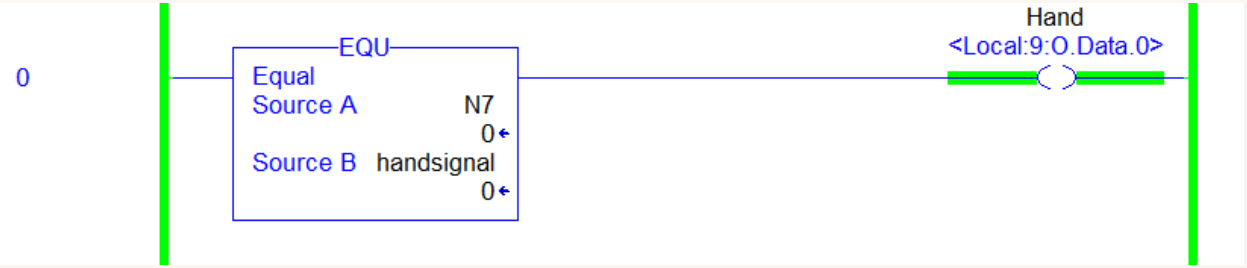
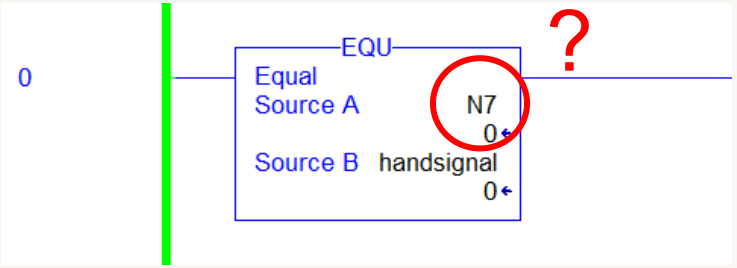


What is the incoming data? Where does it come from?



Data & Signal communication

What are the applicable communication approach in our setup (PC → PLC)

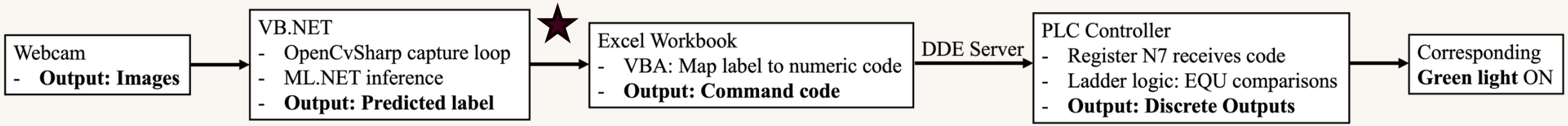
A) Excel + VBA mapping + DDE POKE (RSLinx)

B) OPC (OPC UA or OPC DA via a server)

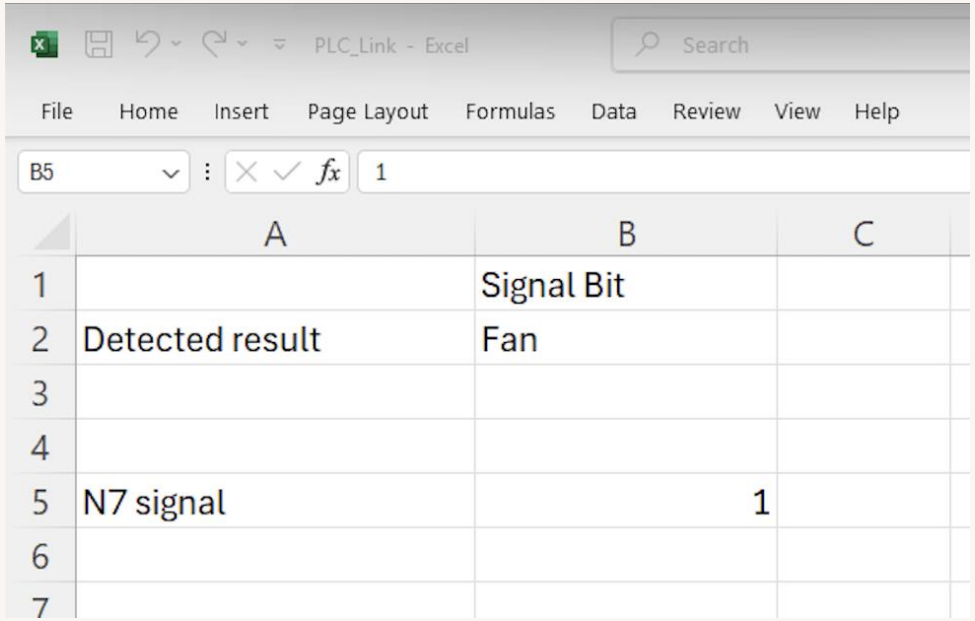
C) HDMI cable from the laptop to the PLC

D) Direct industrial protocol write (e.g., EtherNet/IP / Modbus TCP from the app)

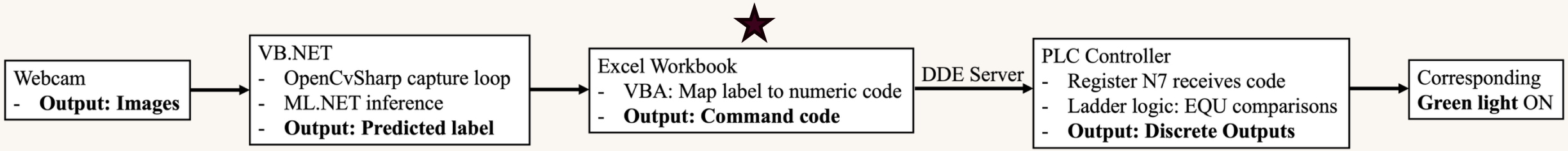
Data & Signal communication



- **VB.NET ML App → Excel**
  - Data: Predicted "LABEL"



Data & Signal communication



- **Excel**

➢ **Translate "LABEL" to integer "CODE"**

Predicted LABEL (from VB)	Integer CODE written to PLC (N7)	Energized Output address	Physical Indicator
DC_motor	0	O:0/0	Green light 0 ON
Fan	1	O:0/1	Green light 1 ON
Motion_sensor	2	O:0/2	Green light 2 ON
LED	3	O:0/3	Green light 3 ON

Data & Signal communication

Webcam
- **Output: Images**

→ VB.NET
- OpenCvSharp capture loop
- ML.NET inference
- **Output: Predicted label**

→ ★ Excel Workbook
- VBA: Map label to numeric code
- **Output: Command code**

→ DDE Server → PLC Controller
- Register N7 receives code
- Ladder logic: EQU comparisons
- **Output: Discrete Outputs**

→ Corresponding **Green light** ON

- **Excel (VBA+Macros)**

➢ **Translate "LABEL" to integer "CODE"**

Predicted LABEL (from VB)	Integer CODE written to PLC (N7)
DC_motor	0
Fan	1
Motion_sensor	2
LED	3

```
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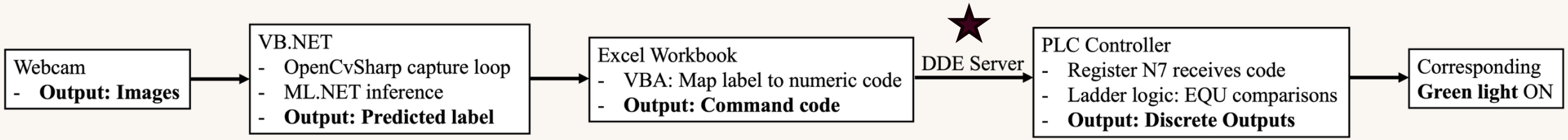
s = LCase$(Trim$(CStr(Worksheets("Sheet1").Range("B2").Value)))
isDCmotor = (s = "dc_motor")
isFan = (s = "fan")
isMotionsensor = (s = "motion_sensor")
isLED = (s = "led")

' 2) Decide B5 values
' write your logic here!!!
If isDCmotor Then
    b5 = 0
ElseIf isFan Then
    b5 = 1
ElseIf isMotionsensor Then
    b5 = 2
ElseIf isLED Then
    b5 = 3
Else
    b5 = 4
End If

' 3) Write to cells (for visibility/debug)
Worksheets("Sheet1").Range("B5").Value = b5
```

Data & Signal communication



- **Excel → RSLinx → PLC**
  - RSLinx: Topic that connect to PLC
  - Native Excel integration via DDE
  - DDE: Dynamic Data Exchange
  - Data: integer "CODE"

```
' 3) Write to cells (for visibility/debug)
Worksheets("Sheet1").Range("B5").Value = b5

' 4) DDE poke both BOOL tags (1/0)
On Error GoTo clean
'Change the "MLproject" below to your topic name
ch = Application.DDEInitiate(app:="RSLinx", TOPIC:="MLproject")
Set ValueToPoke = Worksheets("Sheet1").Range("B5")
Application.DDEPoke ch, "Program:MainProgram.N7", ValueToPoke

Application.DDETerminate ch
Exit Sub
```

Data & Signal communication

Webcam
- **Output: Images**

→

VB.NET
- OpenCvSharp capture loop
- ML.NET inference
- **Output: Predicted label**

→

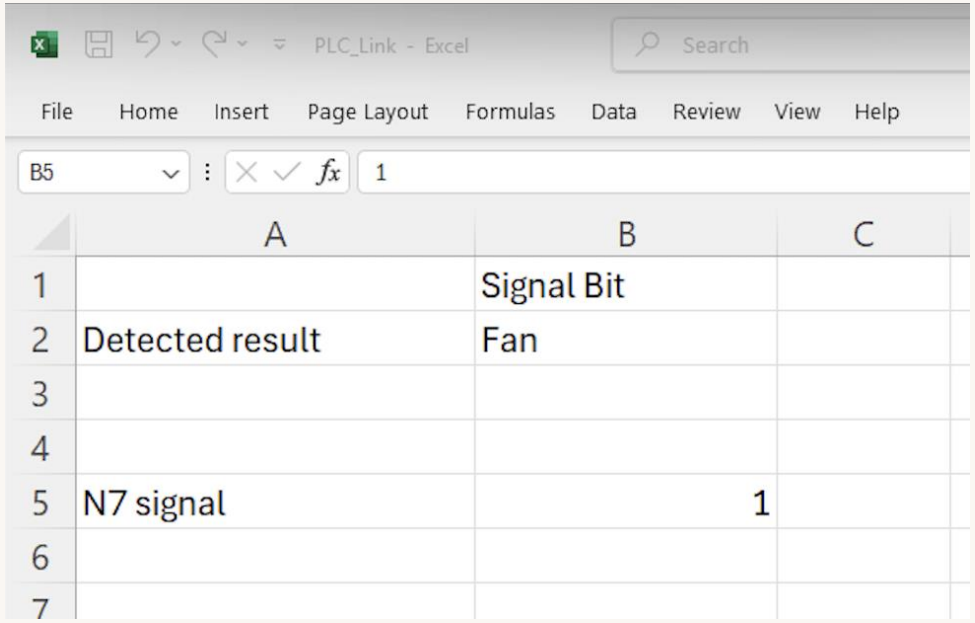
Excel Workbook
- VBA: Map label to numeric code
- **Output: Command code**

→ ★ DDE Server →

PLC Controller
- Register N7 receives code
- Ladder logic: EQU comparisons
- **Output: Discrete Outputs**

→

Corresponding **Green light** ON

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Exit Sub
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Machine Vision (MV)

Webcam
- **Output: Images**

→

VB.NET
- OpenCvSharp capture loop
- ML.NET inference
- **Output: Predicted label**

→

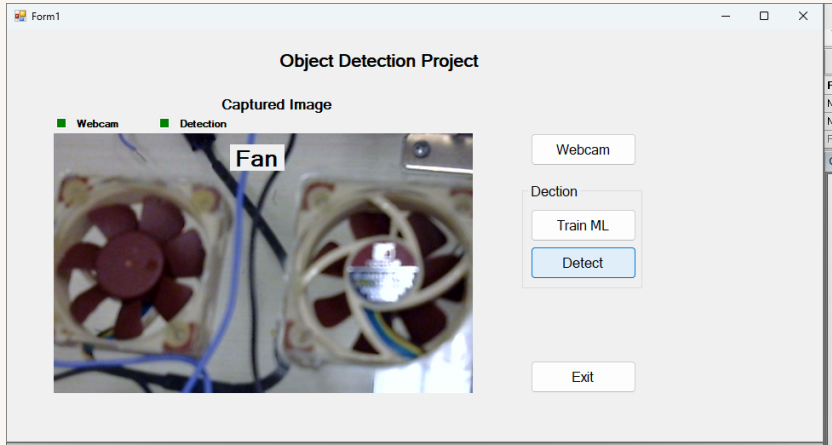
Excel Workbook
- VBA: Map label to numeric code
- **Output: Command code**

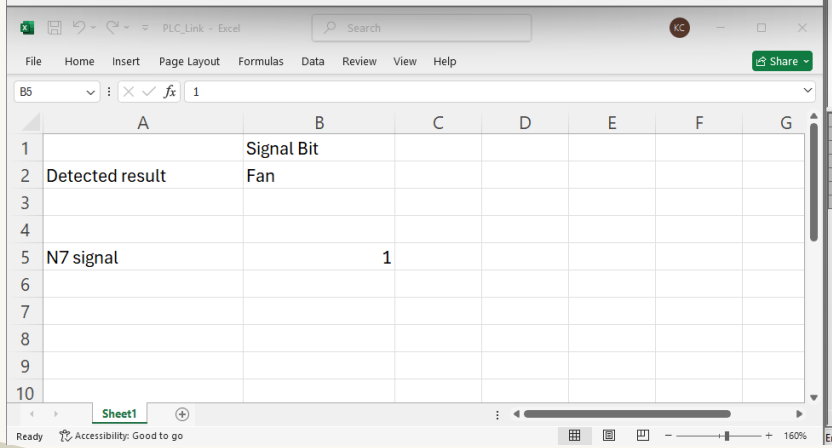
→ DDE Server →

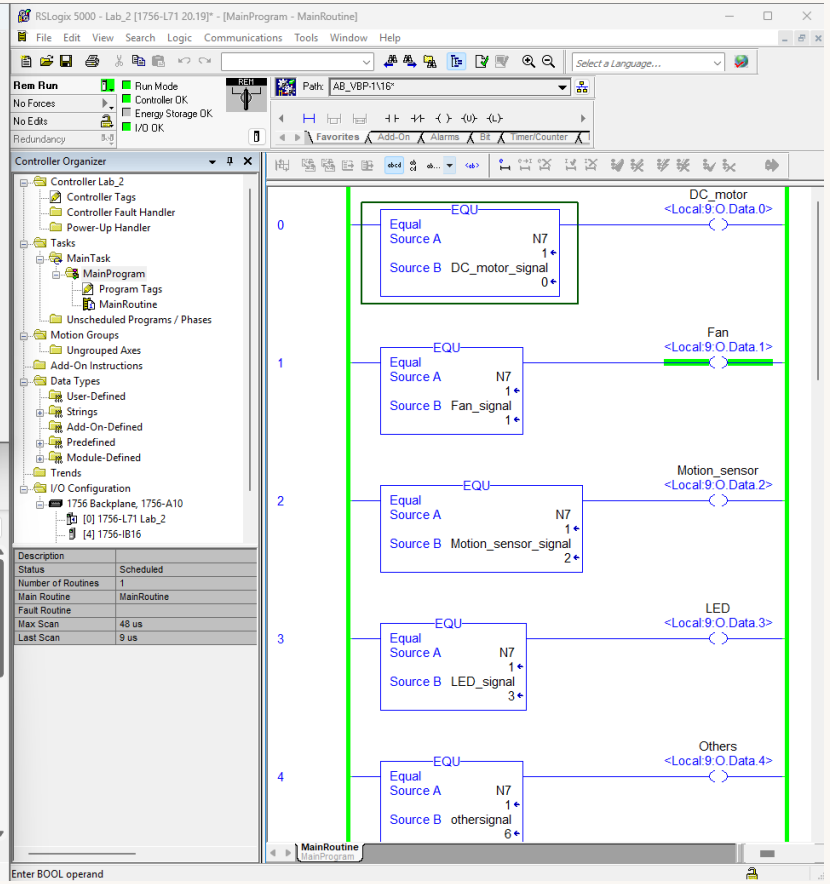
PLC Controller
- Register N7 receives code
- Ladder logic: EQU comparisons
- **Output: Discrete Outputs**

→

Corresponding **Green light** ON







Q4. What type of data are we transmitting (PC → PLC)

A) Image frames

B) Text labels

C) Integer codes

D) Audio signals

Q5. Where does the translation happen (Label to Code) ?

A) Inside the PLC ladder logic

B) In Excel (VBA code)

C) Inside the ML model during training

D) In the ML App for image classification

Q6. What does DDE Poke do in our setup?

A) Reads values from the PLC into Excel

B) Pushes/writes a value from Excel into a RSLinx topic/item

C) Trains the Inception model

D) Converts text labels into integers

Q & A

Lab Demonstration

Lab Outline

- **Equipment setup**
  - Visual basic studio software
  - RSLinx Classic
  - RSLogix 5000
  - Webcam
  - PC
  - PLC

- **Teaching materials**
  - VB.NET App code
  - Excel sheet with VBA code
  - PLC ladder logic diagram
  - Lab handout/instructions

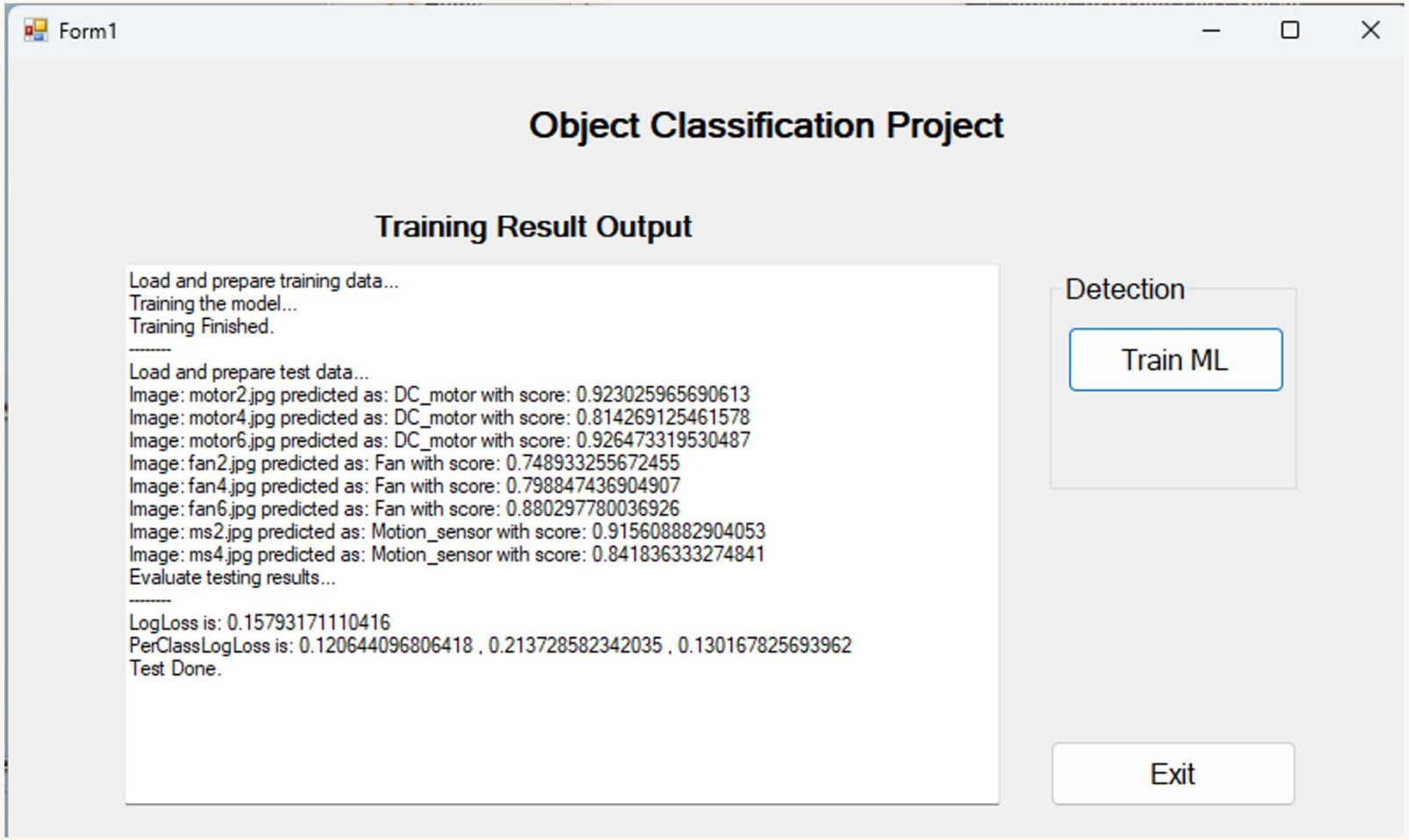
- **Lab 1**
  - Data collection and labeling
  - VB.NET ML App setup and run training

- **Lab 2**
  - VB.NET ML App run testing
  - Real-time deployment with webcam

- **Lab 3**
  - PLC Ladder logic
  - VBA code logic and DDE configuration

Lab 1

- Data collection and labeling
- VB.NET ML App setup and run training



Lab 1

- Data collection and labeling
- VB.NET ML App setup and run training

Lab 1

- Data collection and labeling
- VB.NET ML App setup and run training

Scenario Design

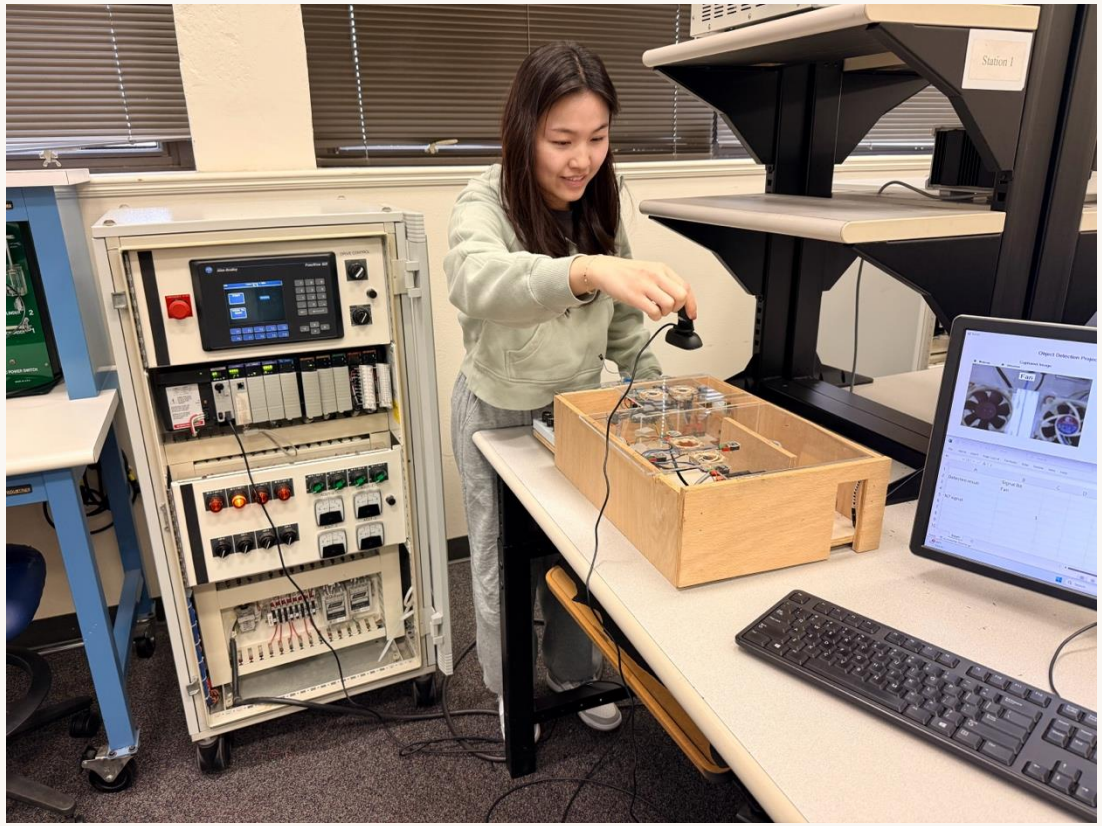
A) Hands Detection

B) Defect Detection

C) Bolt/ Nut/ Washer Classification

Lab 1

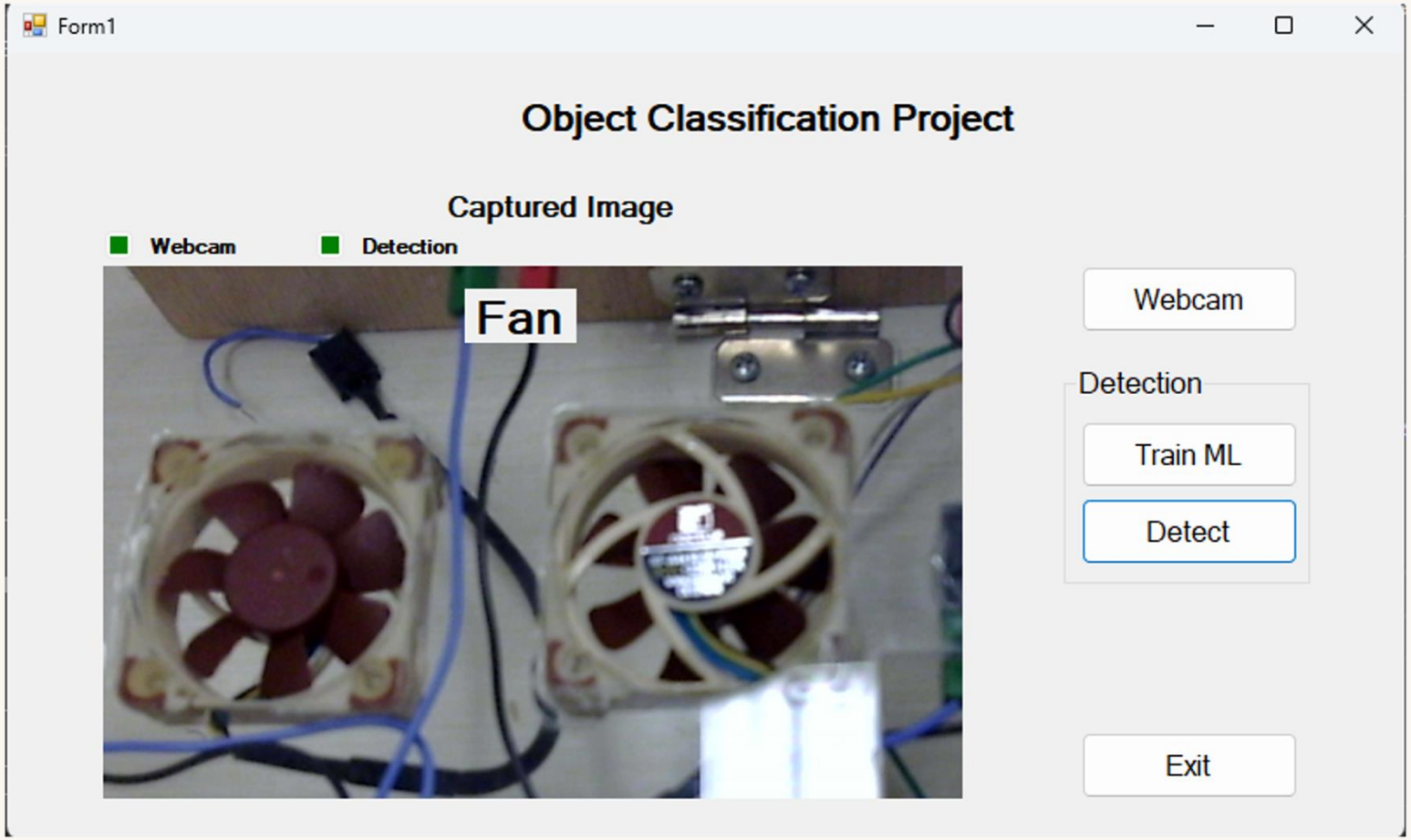
- Data collection and labeling
- VB.NET ML App setup and run training



Let's do it!

Lab 2

- VB.NET ML App run testing
- Real-time deployment with webcam



Lab 2

- VB.NET ML App run testing
- Real-time deployment with webcam

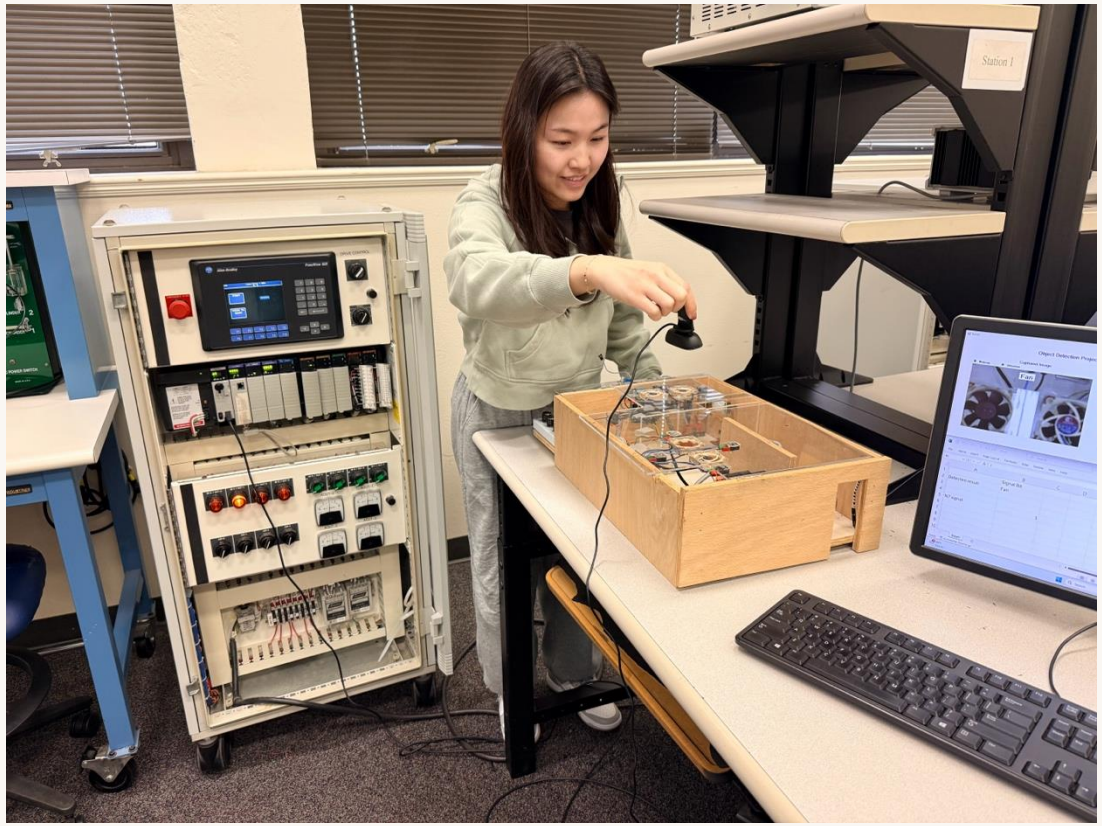
Lab 2

- VB.NET ML App run testing
- Real-time deployment with webcam

What can cause inaccuracy in test/ real-time deployment?

Lab 2

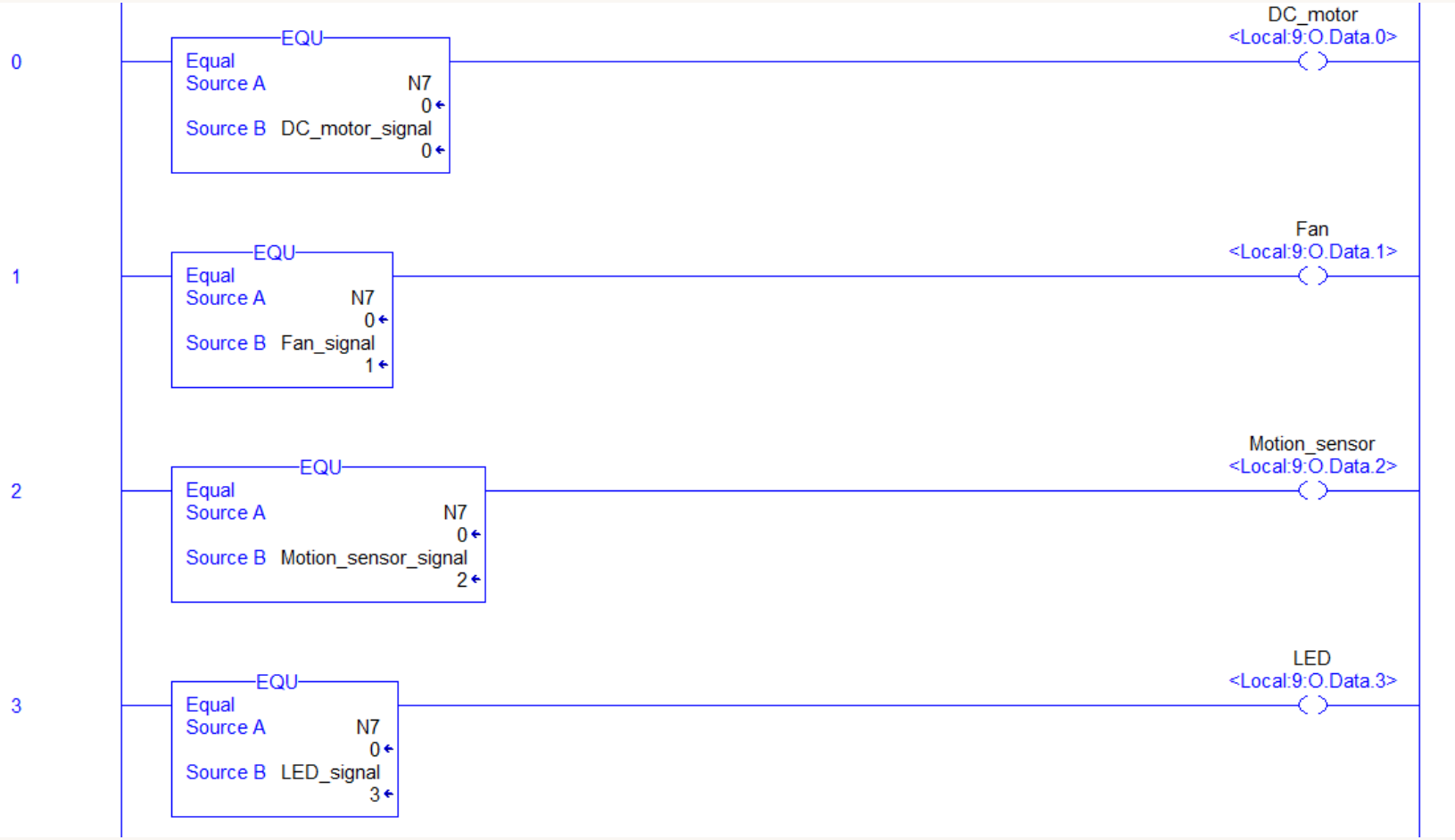
- VB.NET ML App run testing
- Real-time deployment with webcam



Let's do it!

Lab 3

- **PLC Ladder logic**
- VBA code logic and DDE configuration



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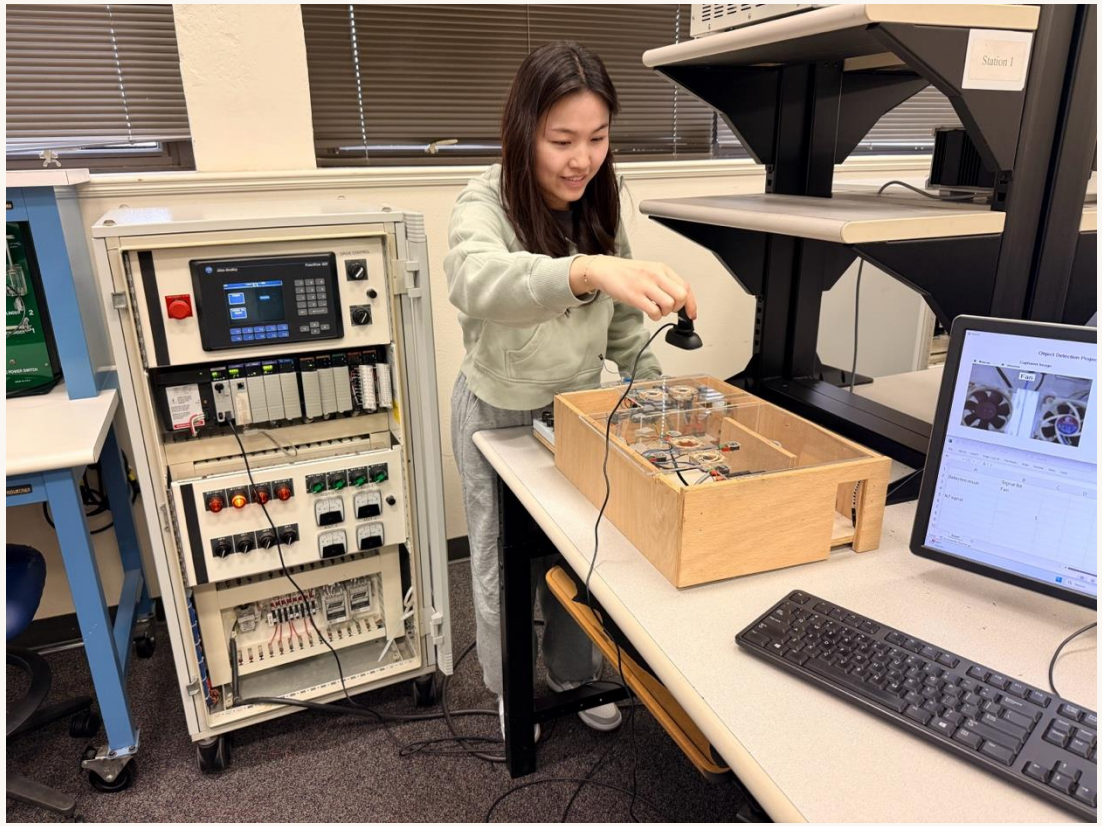

Let's do it!

Lab 3

- **PLC Ladder logic**
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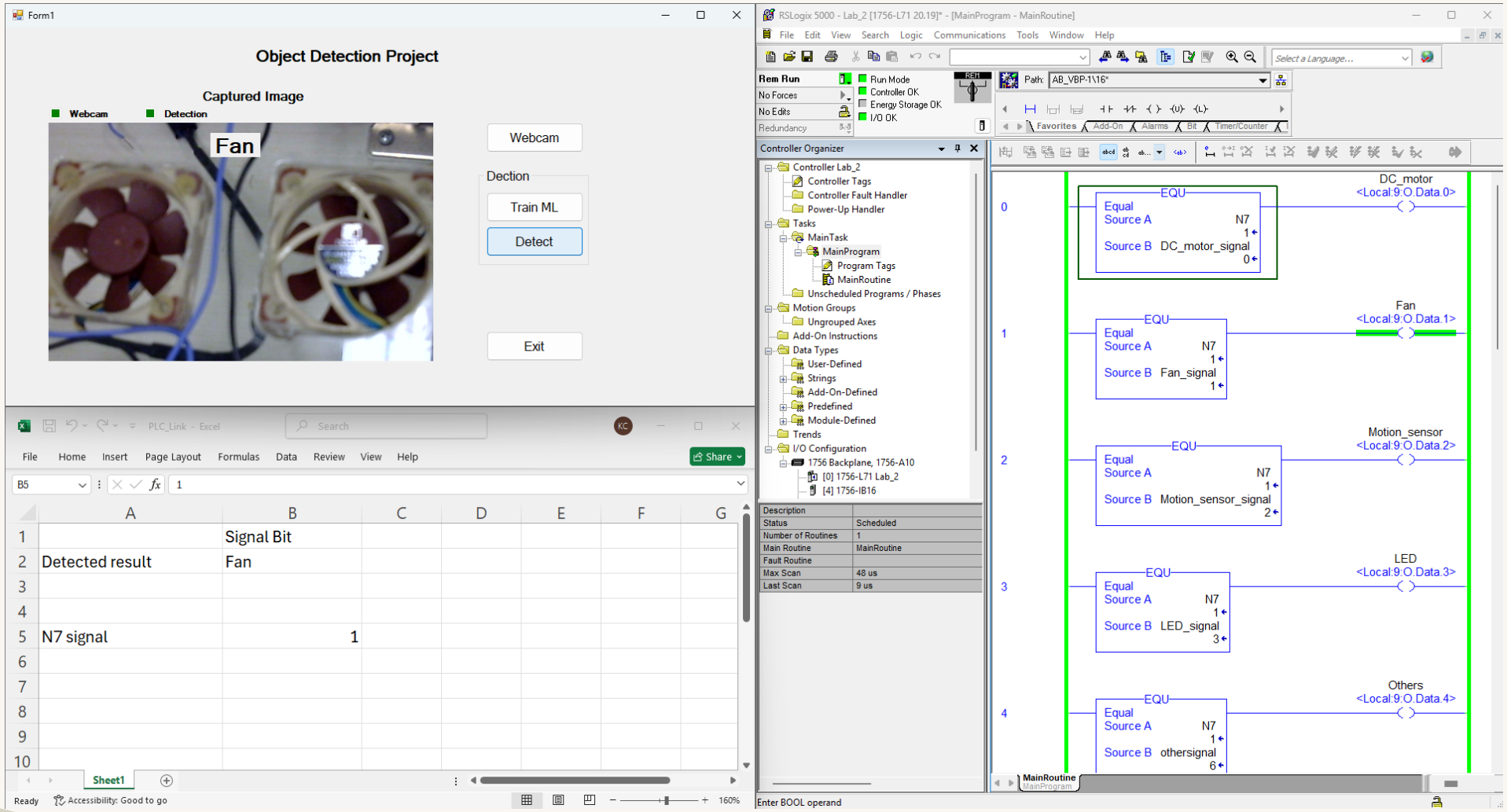
What's next ?

Lab 3

- PLC Ladder logic
- **VBA code logic and DDE configuration**



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ElseIf isMotionsensor Then
    b5 = 2
ElseIf isLED Then
    b5 = 3
Else
    b5 = 4
End If

' 3) Write to cells (for visibility/debug)
Worksheets("Sheet1").Range("B5").Value = b5

' 4) DDE poke both BOOL tags (1/0)
On Error GoTo clean
'Change the "MLproject" below to your topic name
ch = Application.DDEInitiate(app:="RSLinx", TOPIC:="MLproject")
Set ValueToPoke = Worksheets("Sheet1").Range("B5")
Application.DDEPoke ch, "Program:MainProgram.N7", ValueToPoke

Application.DDETerminate ch
Exit Sub
```

A

B

C

D

What needs to be modified?

A) Cell Numbers

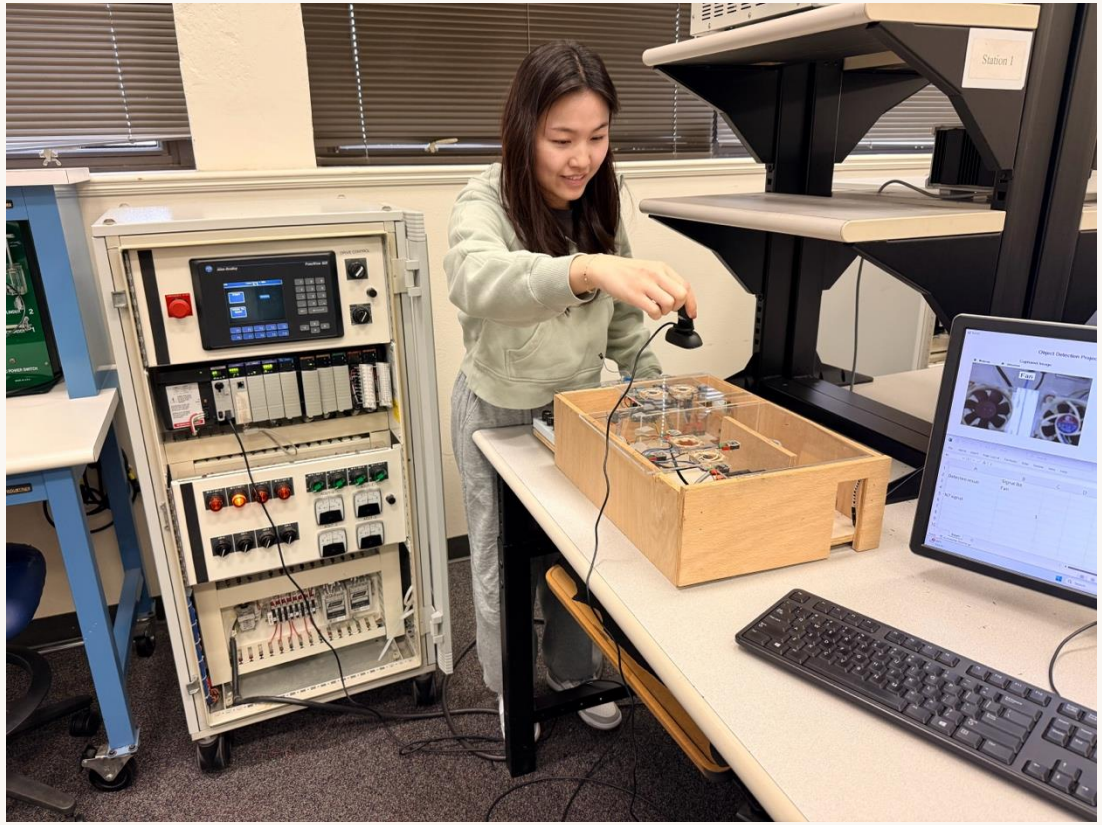
B) Labels in the Boolean flag checking

C) The variables in the “If Then” logic

D) The TOPIC name

Lab 3

- PLC Ladder logic
- **VBA code logic and DDE configuration**



Let's do it!

Q & A

THANK YOU

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